

Integrative Bioinformatics Analysis of hsa-miR-873-5p and Its Predicted Target Genes in Glioblastoma

Independent bioinformatics research report

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1.INTRODUCTION	5
2. RESEARCH QUESTION	5
3. OBJECTIVES.....	5
4.METHODS.....	6
4.1 DATA ACQUISITION AND DATASET DESCRIPTION	6
4.2 SELECTION OF HSA-MIR-873-5P AND MIRNA EXPRESSION ANALYSIS.....	6
4.3 TARGET PREDICTION USING MIRDB	6
4.4 CROSS-VALIDATION OF TARGET PREDICTIONS	6
4.5 PROTEIN–PROTEIN INTERACTION ANALYSIS USING STRING.....	6
4.6 MCL CLUSTERING OF THE STRING NETWORK.....	7
4.7 FUNCTIONAL ENRICHMENT ANALYSIS	7
4.8. LITERATURE REVIEW	7
4.9 INDEPENDENT EXPRESSION ASSESSMENT USING GSE90886.....	7
4.10 STATISTICAL ANALYSIS.....	7
5.RESULTS	8
5.1 MIRNA EXPRESSION ANALYSIS AND SELECTION OF HSA-MIR-873-5P.....	8
5.2 TARGET PREDICTION USING MIRDB	8
5.3 TARGETSCAN/MIRCARTA CROSS-VALIDATION.....	8
5.4 STRING NETWORK AND MCL CLUSTERING	8
5.6 FUNCTIONAL ENRICHMENT ANALYSIS	8
5.7 INDEPENDENT EXPRESSION ASSESSMENT USING GSE90886.....	9
6.DISCUSSION	9
6.1 PRINCIPLE FINDINGS AND INTERPRETATION	9
6.2 BIOLOGICAL RELEVANCE OF THE CANDIDATE TARGETS.....	10
6.3 INTERPRETATION OF THE STRING AND ENRICHMENT FINDINGS.....	10
6.4 INDEPENDENT EXPRESSION ASSESSMENT AND ITS IMPLICATIONS.....	10
6.5 OVERALL INTERPRETATION	11
7.LIMITATIONS	11
7.1 STUDY DESIGN AND SAMPLE SIZE	11

7.2 LIMITATIONS OF COMPUTATIONAL TARGET PREDICTION	11
7.3 LIMITATIONS OF INDEPENDENT EXPRESSION ASSESSMENT	11
8.CONCLUSION	12
9.REFERENCES	12

ABSTRACT

Glioblastoma is the most aggressive primary brain tumour. marked by complex molecular alterations and considerable biological heterogeneity. MicroRNAs regulate gene expression and play roles in cancer progression. This study investigated Hsa-miR-873-5p expression in glioblastoma using GEO dataset GSE103228. Target prediction through miRDB yielded 38 higher-scoring candidates genes. These candidates were then cross-referenced with TargetScan-supported prediction (via miRCarta), yielding 17 overlapping candidate genes. STRING was used to further examine these candidates for protein-protein interaction analysis, functional enrichment, MCL clustering, literature review and independent expression assessment using GEO dataset GSE90886. Hsa-miR-872-5p was found to be markedly downregulated in glioblastoma relative to normal brain tissue (mean:0.0419 vs 1.2151).Independent assessment of the 17 candidates in GSE90886 did not show statistically significant differential expression after multiple-testing correction, and no robust functional enrichment was observed. Overall, these findings provide a hypothesis-generating set of candidate hsa-miR-873-5p targets for further investigation, with experimental validation required to confirm direct regulatory relationships and biological relevance in glioblastoma.

Key words: Glioblastoma; hsa-miR-873-5p; microRNA; bioinformatics; miRDB; TargetScan; gene expression; computational analysis

1.INTRODUCTION

Glioblastoma (GBM) is the most aggressive primary brain tumour, marked by rapid growth, invasiveness and considerable biological heterogeneity. Despite advances in diagnosis and treatment, its complex molecular characteristics and poor clinical outcomes make GBM a challenging disease (Khalafallah et al., 2021; Becker et al., 2021).

MicroRNAs (miRNAs) are small non-coding RNAs that regulate gene expression by targeting messenger RNAs, thereby influencing key cancer-related processes, including cell proliferation, migration, and apoptosis, and their dysregulation has been widely implicated in tumorigenesis (Hwang & Mendell, 2006; Di Leva et al., 2014).

Hsa-miR-873-5p has attracted particular interest due to its reported downregulation in tumour tissue and evidence of tumour-suppressive activity (Zhang et al., 2023). Specific molecular targets of hsa-miR-873-5p have also been experimentally reported in other cancer contexts (Chen & Ji, 2021; Chen et al., 2021). These findings provide a rationale for further characterising its regulatory network in glioblastoma.

Building on this evidence, the present study analysed hsa-miR-873-5p expression in glioblastoma versus normal brain tissue using the GEO dataset GSE103228. Computational target prediction was then performed using miRDB and cross-validated against TargetScan-supported predictions via miRCarta. The resulting candidate targets were further examined using protein-protein interaction analysis, functional enrichment analysis, literature review and independent expression validation. The overall aim was to generate a biologically informed, hypothesis-generating set of potential hsa-miR-873-5p target genes for further investigation in glioblastoma.

2. RESEARCH QUESTION

What is the expression pattern of hsa-miR-873-5p in glioblastoma compared with normal brain tissue, and which potential target genes and molecular associations can be identified through computational analysis and independent expression assessment?

3. OBJECTIVES

1. To analyse the expression of hsa-miR-873-5p in glioblastoma and normal brain tissue using the GEO dataset GSE103228
2. To identify candidate hsa-miR-873-5p targets through miRDB prediction and cross-reference them against TargetScan-supported data obtained via miRCarta
3. To analyse protein-protein interactions and functional clusters using STRING and MCL

4. To evaluate the biological relevance of candidate genes through functional enrichment analysis and literature review.
5. To examine candidate gene expression in the independent GEO dataset GSE90886 as an independent expression assessment.

4.METHODS

4.1 DATA ACQUISITION AND DATASET DESCRIPTION

For the analysis of miR-873-5p, the dataset was obtained from the NCBI Gene Expression Omnibus (GEO) under accession number GSE103228. The dataset contains 10 human brain tissue samples with miRNA expression profiles, including five normal brain samples and five glioblastoma samples. The expression data were generated using the Exiqon miRCURY LNA microRNA Array, 7th generation, corresponding to the GEO platform GPL18058. The processed expression matrix provided by GEO was used for downstream analysis (NCBI GEO, GSE103228).

4.2 SELECTION OF HSA-MIR-873-5P AND MIRNA EXPRESSION ANALYSIS

Expression values from the processed matrix were compared between the five glioblastoma and five normal brain samples. Hsa-miR-873-5p showed lower expression in all five glioblastoma samples compared with the normal samples. Based on this observed expression pattern, hsa-miR-873-5p was selected as the primary miRNA for further analysis.

4.3 TARGET PREDICTION USING MIRDB

Candidate target genes of hsa-miR-873-5p were identified using the miRDB database. Predictions with a Target Score of ≥ 90 were selected to focus the analysis on higher-scoring predicted interactions. This resulted in 38 candidate target genes for hsa-miR-873-5p, which were subsequently compared with TargetScan-supported predictions.

4.4 CROSS-VALIDATION OF TARGET PREDICTIONS

To increase confidence in the computational predictions, the 38 candidate genes obtained from miRDB were cross-referenced with TargetScan-supported target predictions accessed through the miRCarta resource. Genes common to both prediction sets were retained as overlapping candidate targets, resulting in 17 genes for downstream analysis (Agarwal et al., 2015; Backes et al., 2018).

4.5 PROTEIN–PROTEIN INTERACTION ANALYSIS USING STRING

Protein–protein interaction analysis was performed by submitting the 17 overlapping candidate genes to the STRING database (Szklarczyk et al., 2019). The initial network was examined for connectivity and protein–protein interaction enrichment. As no significant enrichment was observed in the initial network, a first-shell network expansion was applied to include STRING-

associated interaction partners. The expanded network was subsequently used for downstream clustering analysis.

4.6 MCL CLUSTERING OF THE STRING NETWORK

To identify groups of closely associated proteins within the expanded STRING network, clustering was performed using the Markov Cluster Algorithm (MCL) (van Dongen, 2000) with an inflation parameter of 2.5. The resulting network was divided into seven clusters, enabling further examination of the functional organisation among the interacting proteins.

4.7 FUNCTIONAL ENRICHMENT ANALYSIS

To investigate the biological functions and pathways associated with the 17 candidate genes, functional enrichment analysis was performed using g:Profiler (Raudvere et al., 2019). Gene Ontology categories, including Biological Process, Molecular Function, and Cellular Component, as well as pathway databases such as KEGG, Reactome, and WikiPathways, were examined. Statistical significance was assessed using multiple-testing correction. The original 17-gene set was analysed separately from the expanded STRING network to distinguish predicted miRNA targets from STRING-associated interaction partners.

4.8. LITERATURE REVIEW

As computational prediction alone does not establish functional relevance, the 17 candidate genes were reviewed against previously published literature for evidence of glioblastoma association and other relevant biological processes. This included searching for evidence of direct miRNA–target interactions, glioblastoma-related functions or expression patterns, and other cancer-related findings. The findings were used to support interpretation of the candidate targets and were not treated as experimental confirmation.

4.9 INDEPENDENT EXPRESSION ASSESSMENT USING GSE90886

The expression behaviour of the 17 candidate genes was independently assessed using the GEO dataset GSE90886 (NCBI GEO, GSE90886), containing nine glioblastoma and nine normal brain tissue samples generated on the Affymetrix Human Gene Expression Array (GPL15207).

Multiple probes corresponding to the same gene were identified using the platform annotation, and probe-level values were summarised within each sample using the median. Gene expression levels were then compared between the glioblastoma and normal brain groups.

4.10 STATISTICAL ANALYSIS

To statistically compare gene expression between the glioblastoma and normal brain groups, Welch’s two-sample t-test was used. Multiple-testing correction was performed using the Benjamini–Hochberg method (Benjamini & Hochberg, 1995), and an adjusted p-value below 0.05 was considered statistically significant. All 17 candidate genes were included in the multiple-testing correction.

5.RESULTS

5.1 MIRNA EXPRESSION ANALYSIS AND SELECTION OF HSA-MIR-873-5P

Analysis of the GSE103228 dataset showed markedly reduced expression of hsa-miR-873-5p in glioblastoma samples compared to normal brain samples. The mean expression was 1.2151 in the normal group and 0.0419 in the glioblastoma group, representing approximately 29-fold lower mean expression in glioblastoma group. This downregulation was consistent across all samples, with each of the five normal samples showing higher expression than every glioblastoma sample. Based on this expression pattern, hsa-miR-873-5p was selected as the miRNA of interest for downstream target prediction and functional investigation.

5.2 TARGET PREDICTION USING MIRDB

Candidate target genes of hsa-miR-873-5p were predicted using miRDB (Chen & Wang, 2020). A total of 38 candidate target genes with a Target Score of ≥ 90 were selected for further cross-validation with TargetScan-supported predictions.

5.3 TARGETSCAN/MIRCARTA CROSS-VALIDATION

Cross-referencing the 38 miRDB candidates with TargetScan-supported predictions (via miRCarta) identified 17 genes common to both sets. These 17 genes were retained as the candidate target set for downstream network, enrichment, literature, and expression analyses.

5.4 STRING NETWORK AND MCL CLUSTERING

The initial STRING analysis network of 17 candidate genes contained no detected interaction edges, with no significant PPI enrichment observed ($p=1$). The network was subsequently expanded using first-shell STRING-associated interaction partners. MCL clustering with an inflation parameter of 2.5 produced seven clusters containing a total of 17 proteins, including seven proteins from the original candidate set and ten additional proteins added during STRING network expansion.

5.6 FUNCTIONAL ENRICHMENT ANALYSIS

Enrichment analysis of the 17 original candidate genes was performed using g:Profiler. No statistically significant enrichment was identified across Gene Ontology categories, including Biological Process, Molecular Function, and Cellular Component, or across the examined pathway databases, including KEGG, Reactome, and WikiPathways. A borderline association with the CORUM NDPKA-AMPK α 1 complex approached the significance threshold (adjusted $p \approx 0.050$) but was treated with caution and was not considered a robust pathway-level finding.

5.7 INDEPENDENT EXPRESSION ASSESSMENT USING GSE90886

Expression of the 17 candidate genes was assessed independently using the GSE90886 dataset, which contains nine glioblastoma and nine normal brain samples. A total of 57 probes corresponding to the 17 candidate genes were identified, and where multiple probes represented the same gene, probe-level values were summarised within each sample using the median. Differential expression was assessed using Welch's two-sample t-test followed by Benjamini-Hochberg correction. None of the 17 candidate genes reached statistical significance after multiple-testing correction (adjusted $p < 0.05$).

Table 1. Independent expression assessment of the 17 candidate genes in GSE90886

Gene	Probes	GBM Mean	Normal Mean	P-value	Adjusted P-value
AFAP1L2	3	5.348	5.403	0.8355	0.9469
ATP8B2	3	4.492	4.409	0.4228	0.9345
DPF2	5	4.653	4.832	0.1420	0.7818
E2F7	2	4.822	4.978	0.7237	0.9345
FAM83A	2	4.618	4.504	0.1374	0.7818
FOXK2	2	7.279	7.337	0.7353	0.9345
GPRASP1	2	5.737	5.556	0.6764	0.9345
KLHL11	1	3.450	3.394	0.5527	0.9345
NPTN	5	7.487	7.518	0.9561	0.9561
PCSK5	2	6.346	6.566	0.6261	0.9345
PRKAA1	4	3.762	3.710	0.7696	0.9345
PSD3	5	8.716	8.221	0.3601	0.9345
SERTAD2	2	5.603	5.761	0.1840	0.7818
SRRM2	5	7.641	7.612	0.9213	0.9561
THUMPD1	3	8.763	8.556	0.1068	0.7818
TNRC6B	10	7.821	7.732	0.6372	0.9345
ZSWIM5	1	5.987	6.108	0.6266	0.9345

6.DISCUSSION

6.1 PRINCIPLE FINDINGS AND INTERPRETATION

Hsa-miR-873-5p was identified as strongly downregulated in the GSE103228 glioblastoma dataset. Computational prediction using miRDB generated 38 higher-scoring candidate target genes, 17 of which were also supported by TargetScan-derived predictions via miRCarta and retained as the final candidate set. The lack of statistically significant differential expression of these candidate genes in the independent GSE90886 dataset does not provide independent

expression supports the interpretation that for the candidate set; the genes should therefore be regarded as hypotheses rather than confirmed glioblastoma-specific regulatory targets.

6.2 BIOLOGICAL RELEVANCE OF THE CANDIDATE TARGETS

Several predicted target genes have previously been investigated in glioblastoma or related glioma contexts. E2F7 has been investigated in glioblastoma, where it was reported to be upregulated and to promote glioblastoma cell proliferation, metastasis, and tumorigenicity (Yang et al., 2020). PCSK5 has also been investigated in glioblastoma, including in relation to STAT3-associated tumour biology (Gong et al., 2025). FOXK2 has been studied in glioma, where its expression was reported to decrease with increasing tumour grade and its experimental modulation affected glioma cell proliferation, migration, and invasion (Wang et al., 2018). PSD3 has been identified in studies of astrocytoma progression, including comparisons involving secondary glioblastoma (van den Boom et al., 2006). ZSWIM5 has also been reported among genes with strong expression in high-grade glioma tissue (Meyer, 2014).

Additionally, direct regulatory relationships between hsa-miR-873-5p and PRKAA1 or THUMPD1 have been experimentally reported in gastric cancer (Chen & Ji, 2021; Chen et al., 2021). These findings support the biological plausibility of some candidate interactions but do not confirm direct regulation by hsa-miR-873-5p in glioblastoma.

6.3 INTERPRETATION OF THE STRING AND ENRICHMENT FINDINGS

In the initial STRING network, no significant protein–protein interaction (PPI) enrichment was observed among the 17 candidate targets, suggesting that these genes did not form a strongly connected interaction network under the selected conditions.

Network expansion introduced additional STRING-associated proteins and enabled the identification of seven MCL clusters. However, because these added proteins were not part of the original predicted target set, the expanded network should be interpreted as providing functional context rather than direct evidence of hsa-miR-873-5p target regulation.

Similarly, functional enrichment analysis of the original 17 candidate genes did not identify significant enrichment for Gene Ontology categories or the examined KEGG, Reactome, or WikiPathways pathways. Therefore, no clearly overrepresented biological pathway could be established for the original candidate set.

6.4 INDEPENDENT EXPRESSION ASSESSMENT AND ITS IMPLICATIONS

No statistically significant differential expression was observed for any of the 17 candidate genes in the independent GSE90886 dataset, representing an important limitation of the present findings. This lack of independent expression support does not necessarily rule out biological regulation, as miRNA–target relationships can be context-dependent and may not be detectable in bulk

expression data. Nonetheless, the absence of independent support limits the strength of the conclusions that can be drawn from the present analysis.

6.5 OVERALL INTERPRETATION

Overall, this analysis provides a computational framework for identifying potential targets of hsa-miR-873-5p in glioblastoma. The consistent downregulation of hsa-miR-873-5p in GSE103228, together with 17 overlapping computational candidate targets, provides a foundation for further investigation. However, the absence of significant functional enrichment and independent expression support means that these findings should be regarded as hypothesis-generating rather than confirmatory. Experimental validation, including direct miRNA–target assays and gene expression or functional studies in relevant glioblastoma models, would be required to establish specific regulatory relationships.

7. LIMITATIONS

7.1 STUDY DESIGN AND SAMPLE SIZE

A key limitation of this study is the relatively small sample size in both datasets. The discovery dataset GSE103228 comprised five normal and five glioblastoma samples, while the independent expression assessment dataset GSE90886 comprised nine normal and nine glioblastoma samples. These small sample sizes may limit statistical power and may not fully capture the biological heterogeneity of glioblastoma. Accordingly, the findings should be considered exploratory and warrant further assessment in larger, independently collected cohorts.

7.2 LIMITATIONS OF COMPUTATIONAL TARGET PREDICTION

The target genes identified in this study were predicted using computational resources and therefore cannot be considered experimentally confirmed miRNA targets. Cross-referencing miRDB and TargetScan-supported predictions increased confidence in the candidate set by identifying genes supported by both resources. However, computational prediction does not eliminate the possibility of false-positive or context-dependent interactions. Similarly, overlap between prediction resources does not demonstrate that hsa-miR-873-5p directly regulates these genes in glioblastoma. Experimental approaches such as luciferase reporter assays, miRNA overexpression or inhibition, and gene expression or functional studies would be required to confirm direct regulatory relationships.

7.3 LIMITATIONS OF INDEPENDENT EXPRESSION ASSESSMENT

The independent expression assessment relied on a publicly available bulk-tissue dataset, which may contain biological and cellular heterogeneity capable of influencing gene-expression measurements. The absence of statistically significant differential expression among the 17 candidate genes does not necessarily exclude genuine regulatory relationships, as miRNA–target interactions may be tissue-, cell-type-, or context-dependent. Additionally, the analysis used

processed expression values provided by GEO rather than independently reprocessed raw data. Therefore, the GSE90886 findings should be interpreted as an independent expression assessment rather than definitive validation.

8.CONCLUSION

This study applied a stepwise computational pipeline to identify potential targets of hsa-miR-873-5p in glioblastoma. Expression analysis of GSE103228 demonstrated marked downregulation of hsa-miR-873-5p in glioblastoma relative to normal brain tissue. This was followed by computational target prediction and cross-database comparison, which identified 17 candidate target genes. Network and functional enrichment analyses were then used to provide additional biological context, while independent expression assessment using GSE90886 did not reveal statistically significant differential expression of the candidate genes following multiple-testing correction.

Collectively, these findings provide a hypothesis-generating set of potential hsa-miR-873-5p target genes that may warrant further investigation. However, the computational predictions and independent expression assessment do not establish direct regulatory relationships in glioblastoma. Experimental studies, including direct miRNA–target assays and functional investigations in relevant glioblastoma models, are required to determine whether these candidate interactions have biological significance in glioblastoma.

9.REFERENCES

- Agarwal V, Bell GW, Nam JW, Bartel DP. Predicting effective microRNA target sites in mammalian mRNAs. *eLife*. 2015;4:e05005. doi:10.7554/eLife.05005.
- Backes C, Fehlmann T, Kern F, Kehl T, Lenhof HP, Meese E, Keller A. miRCarta: a central repository for collecting miRNA candidates. *Nucleic Acids Research*. 2018;46(D1):D160-D167. doi:10.1093/nar/gkx851.
- Becker AP, Sells BE, Haque SJ, Chakravarti A. Tumor heterogeneity in glioblastomas: from light microscopy to molecular pathology. *Cancers (Basel)*. 2021;13(4):761. doi:10.3390/cancers13040761.
- Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*. 1995;57(1):289-300. doi:10.1111/j.2517-6161.1995.tb02031.x.
- Chen Q, Lin L, Xiong B, Yang W, Huang J, Shi H, Wang Z. MiR-873-5p targets THUMP1 to inhibit gastric cancer cell behavior and chemoresistance. *Journal of Gastrointestinal Oncology*. 2021;12(5):2061-2072. doi:10.21037/jgo-21-641.

Chen W, Ji Y. CircC6orf132 facilitates proliferation, migration, invasion, and glycolysis of gastric cancer cells under hypoxia by acting on the miR-873-5p/PRKAA1 axis. *Frontiers in Genetics*. 2021;12:636392. doi:10.3389/fgene.2021.636392.

Chen Y, Wang X. miRDB: an online database for prediction of functional microRNA targets. *Nucleic Acids Research*. 2020;48(D1):D127-D131. doi:10.1093/nar/gkz757.

Di Leva G, Garofalo M, Croce CM. MicroRNAs in cancer. *Annual Review of Pathology: Mechanisms of Disease*. 2014;9:287-314. doi:10.1146/annurev-pathol-012513-104715.

Gong H, Yang X, An L, Zhang W, Liu X, Shu L, Yang L. PCSK5 downregulation promotes the inhibitory effect of andrographolide on glioblastoma through regulating STAT3. *Molecular and Cellular Biochemistry*. 2025;480(1):521-533. doi:10.1007/s11010-024-04977-3.

Hwang HW, Mendell JT. MicroRNAs in cell proliferation, cell death, and tumorigenesis. *British Journal of Cancer*. 2006;94(6):776-780. doi:10.1038/sj.bjc.6603023.

Khalafallah A, Huq S, Jimenez AE, Serra R, Bettegowda C, Mukherjee D. “Zooming in” on glioblastoma: understanding tumor heterogeneity and its clinical implications in the era of single-cell ribonucleic acid sequencing. *Neurosurgery*. 2021;88(3):477-486. doi:10.1093/neuros/nyaa305.

Meyer MA. Highly expressed genes in human high grade gliomas: immunohistochemical analysis of data from the Human Protein Atlas. *Neurology International*. 2014;6(2):5348. doi:10.4081/ni.2014.5348.

National Center for Biotechnology Information. Gene Expression Omnibus (GEO): GSE103228. Bethesda (MD): National Library of Medicine. Available from: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE103228>.

National Center for Biotechnology Information. Gene Expression Omnibus (GEO): GSE90886. Bethesda (MD): National Library of Medicine. Available from: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE90886>.

Raudvere U, Kolberg L, Kuzmin I, Arak T, Adler P, Peterson H, Vilo J. G:Profiler: a web server for functional enrichment analysis and conversions of gene lists (2019 update). *Nucleic Acids Research*. 2019;47(W1):W191-W198. doi:10.1093/nar/gkz369.

Szklarczyk D, Gable AL, Lyon D, Junge A, Wyder S, Huerta-Cepas J, et al. STRING v11: protein-protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Research*. 2019;47(D1):D607-D613. doi:10.1093/nar/gky1131.

van den Boom J, Wolter M, Blaschke B, Knobbe CB, Reifenberger G. Identification of novel genes associated with astrocytoma progression using suppression subtractive hybridization and real-time

reverse transcription-polymerase chain reaction. *International Journal of Cancer*. 2006;119(10):2330-2338. doi:10.1002/ijc.22108.

van Dongen S. *Graph Clustering by Flow Simulation* [dissertation]. Utrecht: University of Utrecht; 2000.

Wang B, Zhang XB, Wang W, Zhu ZZ, Tang F, Wang D, Liu X, Zhuang H, Yan XL. Forkhead box K2 inhibits the proliferation, migration, and invasion of human glioma cells and predicts a favorable prognosis. *OncoTargets and Therapy*. 2018;11:1067-1075. doi:10.2147/OTT.S157126.

Yang R, Wang M, Zhang G, Bao Y, Wu Y, Li X, et al. E2F7-EZH2 axis regulates PTEN/AKT/mTOR signalling and glioblastoma progression. *British Journal of Cancer*. 2020;123:1445-1455. doi:10.1038/s41416-020-01032-y.

Zhang X, Jing F, Guo C, Li X, Li J, Liang G. Tumor-suppressive function and mechanism of miR-873-5p in glioblastoma: evidence based on bioinformatics analysis and experimental validation. *Aging (Albany NY)*. 2023;15(12):5412-5425. doi:10.18632/aging.204800.